

Functional Requirements Specification (FRS)

Subsidy Delivery Partner (Pod Delivery App) — Version 1.2

1. Document Control

Application Name	Subsidy Delivery Partner (Pod Delivery)
Document Version	1.2
Prepared / Developed By	Lokesh Pawalia and Sarthak Srivastava
Prepared Date	August 31, 2026
Reviewed & Approved By	Lokesh Pawalia & Sarthak Srivastava
Document Status	Approved & Production Ready
Live Cloud Backend	Railway Production: https://pod-app-production-818a.up.railway.app
Database & Storage	MongoDB Atlas (pody_db) Cloudinary Media CDN

2. Purpose & System Scope

Purpose: This specification defines the functional requirements for the Subsidy Delivery Partner (Pod Delivery) mobile platform and its cloud backend. Version 1.2 enforces tamper-proof delivery tracking, preventing subsidy diversion and identity impersonation through multi-factor validation.

In Scope: Partner authentication, mandatory 8-step profile onboarding with automatic postal PIN lookup, 600m geofence validation, on-device Google ML Kit facial verification, multi-angle item proof capture, video audit recording, automated PDF invoicing, and native in-app inspection.

Out of Scope: Direct public farmer login portal and commercial payment gateway processing (focus is strictly on physical subsidy handover verification).

3. User Roles & Hierarchy

Role ID	User Role	Description	System Permissions
UR-01	Delivery Partner	Primary field operative delivering subsidy packages.	Login, complete profile, view assigned deliveries, execute POD verification gates.
UR-02	Beneficiary (Farmer)	Registered recipient entitled to subsidy items.	Provides delivery OTP, physical face match, receives subsidized items.
UR-03	System Administrator	Central supervisor monitoring operations and audit logs.	Inspect MongoDB (pody_db), review Cloudinary media proofs, view delivery logs.

4. Functional Requirements Matrix

4.1 Partner Authentication & Strict 8-Step Profile Onboarding

Req ID	Field / Step	Validation & Software Behavior Rule	Priority
FR-PROF-01	1. Full Name	Alphabetical and space characters only via <code>FilteringTextInputFormatter.allow(RegExp(r'[a-zA-Z\s]*'))</code> . Digits & special characters are blocked. Length: 2-50 chars.	P0
FR-PROF-02	2. Email Address	Validates RFC email regex <code>^[w-\.]+\@([w-]+\.)+[w-]{2,4}\$</code> . Whitespace characters are blocked.	P0
FR-PROF-03	3. PIN Code	6 numeric digits. Entering the 6th digit auto-triggers an asynchronous lookup to the Indian Postal API (https://api.postalpincode.in/pincode/{PIN}).	P0
FR-PROF-04	4. City & State	Automatically populated from resolved PIN code. Fields are read-only with a manual edit/unlock toggle for offline overrides.	P0
FR-PROF-05	5. Address	Detailed street / locality address. Mandatory minimum length of 5 characters.	P0
FR-PROF-06	6. Vehicle Type	Structured selection dropdown: Two Wheeler, Three Wheeler, Four Wheeler, or Other.	P0
FR-PROF-07	7. Vehicle Number	Auto-capitalized (<code>UpperCaseTextFormatter</code>). Enforces Indian registration format (e.g. DL 01 AB 1234 or BH Series).	P0
FR-PROF-08	8. Aadhaar Number	Exactly 12 numeric digits adhering to UIDAI specification (cannot start with 0 or 1: <code>^[2-9][0-9]{11}\$</code>).	P0
FR-PROF-09	Profile Photo	Mandatory facial photo capture; streamed via multipart/form-data to <code>PATCH /auth/profile</code> and Cloudinary.	P0

4.2 Order Completion & 5 Verification Gates

Gate	Verification Stage	Technical Mechanism & Enforcement	Priority
Gate 1	600m Geofencing	Haversine distance calculation using GPS coordinates. Blocks submission if operative is >600m away.	P0
Gate 2	Beneficiary OTP	Recipient enters unique 4-digit SMS delivery OTP. Handover cannot proceed without valid match.	P0
Gate 3	On-Device Face Match	Google ML Kit detects facial landmarks from live camera and compares with registered farmer photo.	P0
Gate 4	Photo & Video Proof	Operative captures 1 to 5 delivery item photos and an optional 30-second live handover video audit.	P0
Gate 5	PDF Invoicing & Sync	Generates PDF invoice, uploads to Cloudinary (POD-App/invoices), marks order delivered in MongoDB.	P0

5. Cloud Backend API Specification Matrix

Base URL: <https://pod-app-production-818a.up.railway.app> | Framework: FastAPI + MongoDB

Method	Endpoint	Purpose	Payload	Auth
POST	<code>/auth/check-phone</code>	Check if partner phone exists	JSON	Public
POST	<code>/auth/send-otp</code>	Dispatch SMS OTP to mobile	JSON	Public
POST	<code>/auth/verify-otp</code>	Verify 4-digit mobile OTP	JSON	Public
POST	<code>/auth/login-or-register</code>	Issue JWT bearer access token	JSON	Public
PATCH	<code>/auth/profile</code>	Update 8-step profile & photo	Multipart	Bearer
GET	<code>/auth/me</code>	Fetch authenticated partner info	None	Bearer
GET	<code>/farmers</code>	List assigned deliveries	Query Params	Bearer

POST	/farmers	Register new farmer beneficiary	Multipart	Bearer
POST	/farmers/{id}/upload_proof	Submit POD proofs, video & invoice	Multipart	Bearer

6. Document Approvals & Sign-Off

Project Role	Name	Sign-Off Status	Approval Date
Lead Developer & Architect	Lokesh Pawalia	Approved & Verified	August 31, 2026
Lead Developer & Backend Specialist	Sarthak Srivastava	Approved & Verified	August 31, 2026
Release Engineering	Version 1.2 Production Release	Production Ready	August 31, 2026